Cyber-Physical Architecture and Theoretical Frameworks for Capacitive E-Textile Drone Interfaces

Introduction to the Cyber-Physical Ecosystem

The evolution of human-computer interaction has consistently moved toward reducing the friction between user intent and machine execution, transitioning from command-line interfaces to direct graphical manipulation, and currently advancing into mixed reality and cyber-physical systems[1]. This report delineates the exhaustive theoretical frameworks and practical engineering specifications required to realize a highly specialized performance architecture: a lightweight, fully integrated fabric suit designed to control a swarm of thirteen holographic micro-drones. The ultimate application of this system is the live performance of electronic music, utilizing a custom digital audio workstation (DAW) based on the Mixxx architecture, interfaced entirely through augmented reality (AR) glasses and localized drone manipulation.

Bridging the gap between this ultimate vision and physical reality requires a localized, high-fidelity prototype. The immediate developmental phase concentrates on the engineering of a single capacitive-sensing glove and the corresponding stabilization of a single micro-drone. The control mechanism eschews traditional optical tracking or mechanical joysticks, relying instead on the manipulation of electromagnetic fields—the exact physical principle utilized by the theremin[2]. The goal is to constrain the drone's flight dynamics such that it hovers reactively in the proximity of the performer's hand, establishing a tangible, spatial connection between human motion and machine response.

This architecture requires the synthesis of disparate scientific disciplines. The hardware relies on high-resolution capacitance-to-digital converters (CDCs) woven into conductive crocheted textiles[5]. The algorithmic interpretation of this hardware demands formal frameworks: quantum field theory defines the shifting boundary conditions of the electromagnetic field; information theory maximizes signal integrity across overlapping sensor arrays; set theory mathematically maps human gestures to aerodynamic constraints; systems engineering dictates the physical topology of the components; and cybernetics—specifically second-order cybernetics—defines the continuous, structurally coupled feedback loop between the human performer, the AR visual field, and the autonomous drone.

Theoretical Frameworks for Human-Machine Interfacing

To achieve localized, ultra-low-latency drone control via proximity-based hand movements, the underlying physical and mathematical data structures must be rigorously defined. A simplified inverse-square approximation of capacitance is highly insufficient for the noisy, dynamic environment of a live musical performance. The following frameworks govern the operation of the sensor acquisition and control pipelines.

Quantum Field Theory and Capacitive Boundary Conditions

The fundamental mechanism of capacitive sensing relies on measuring the alterations in an electric field between a primary antenna and a grounded object[2]. The human body, fundamentally a volume of salt water surrounded by a dielectric layer of skin, acts as a highly effective conductor of electricity[3]. The performer is capacitively coupled to the Earth's macroscopic ground, meaning that ions within the body and the Earth shift to create temporary electric currents, allowing the human hand to act as the secondary plate of a capacitor[3]. While the capacitance between a human and the Earth is typically on the order of $100\ \mathrm{pF}$, the mutual capacitance between the human hand and a small metal antenna is on the order of a single picofarad ($1\ \mathrm{pF}$), decreasing inversely with distance[3].

From the advanced perspective of Quantum Field Theory (QFT) and macroscopic quantum electrodynamics, the presence of a biological conductor dynamically alters the vacuum expectation value of the surrounding electromagnetic field[7]. In this architecture, the crocheted conductive yarn nodes act as primary boundary conditions constraining the zero-point fluctuations of the field[7]. While the well-documented Casimir effect describes mechanical forces arising from these quantized boundary conditions at microscopic scales—often modeled through zeta-functional regularization and heat kernel expansion[8]—at the macroscopic scale of the glove sensor, the proximity of the human hand alters the environmental impedance of the system, explicitly modifying the driving-point input admittance $Y_{in}(\omega)$ [10].

The sensing circuit utilizes an LC tank resonator architecture. By continuously forcing a narrow-band oscillation, the circuit establishes a defined high-frequency electromagnetic field, typically ranging from $10\ \mathrm{kHz}$ to $10\ \mathrm{MHz}$, around the yarn node[11]. The dressed mode frequencies of the entire cyber-physical system are determined by the roots of the boundary condition equation:

$$sY_{in}(s) + 1/L_J = 0$$

where L_J represents the baseline inductance of the tank circuit[10]. As the performer's hand approaches the node, the equivalent capacitance C_{eq} shifts. Because the sensing circuit adds electrical charge to the antenna at a constant, fixed rate, the voltage rise time changes, subsequently shifting the resonant frequency f_{res} [4]:

$$f_{res} = \frac{1}{2\pi\sqrt{L(C_{tank} + \Delta C_{hand})}}$$

By treating the biological hand as a dynamic boundary condition that suppresses high-frequency environmental modes, the system inherently regularizes ultraviolet divergence in the signal[10]. This provides a highly stable, noise-immune measurement of ΔC_{hand}, isolating the intentional movement of the performer from ambient electromagnetic interference.

Information Theory and Signal Integrity

A persistent, significant challenge in the design of electronic textiles is capacitive and inductive crosstalk between adjacent conductive traces[13]. With four sensor nodes packed tightly into the glove (e.g., palm, thumb, index, pinky) and three distributed sequentially up the arm (wrist, elbow, shoulder), the generated electric fields will inevitably overlap. Information theory provides the definitive mathematical tools to extract the true spatial position of the hand from a noisy, highly correlated dataset.

Let the set of digital sensor readings be defined as a vector $\mathbf{S} = [S_1, S_2, S_3, S_4, S_5, S_6, S_7]^T$, representing the continuous capacitance values from the seven nodes. Let the true, physical state of the performer's hand and arm relative to the drone be a continuous random variable $\mathbf{H}$. The mutual information $I(\mathbf{S}; \mathbf{H})$ quantifies the absolute reduction in uncertainty regarding the hand's position given the observed sensor readings[1]:

$$I(\mathbf{S}; \mathbf{H}) = H(\mathbf{H}) - H(\mathbf{H}|\mathbf{S})$$

where $H(\mathbf{H})$ represents the Shannon entropy of the hand's possible physical states. Because adjacent crocheted nodes—such as the wrist and the palm—will inevitably register correlated signal fluctuations from a single anatomical movement, the joint entropy $H(S_i, S_j)$ is strictly less than the sum of their individual marginal entropies.

To maximize the channel capacity of the glove interface, the control algorithm must implement a robust decorrelation matrix. Techniques such as Principal Component Analysis (PCA) or Independent Component Analysis (ICA) are applied to orthogonalize the raw sensor data before it is mapped to the drone's flight controls. This rigorous information-theoretic filtering ensures that a singular physical movement—such as extending the index finger—translates into an isolated, high-confidence control command, rather than a smeared, multi-node activation that would destabilize the drone.

Set Theory and State-Space Mapping

Set theory provides the formal logic to define the operational state-spaces and mapping functions required for the systems engineering of the control loop. The entire cyber-physical performance system can be defined as a universal set U, comprising the performer P, the sensor glove G, the micro-drone D, and the augmented reality interface A.
We define the topological space of the localized glove sensors as $\mathcal{S} = \{n_1, n_2, n_3, n_4, n_5, n_6, n_7\}$, where specific subsets of $\mathcal{S}$ correspond to localized limb controls. The power set $\mathcal{P}(\mathcal{S})$ contains all possible permutations of sensor node activations. A human gesture g is defined as an element of the continuous temporal evolution of $\mathcal{S}$, such that $g \in \mathbb{R}^7 \times \mathbb{T}$.
The drone's kinematic state space is defined as $\mathcal{K} \subset \mathbb{R}^3 \times SO(3)$, representing its three-dimensional position and orientation. A surjective mapping function M must be engineered to bridge the human and machine states:

$$M : \mathcal{G}_{valid} \to \mathcal{K}_{safe}$$

where $\mathcal{G}_{valid} \subset \mathbb{R}^7 \times \mathbb{T}$ represents the subset of intentional, calibrated human gestures within the bounds of physical anatomy, and $\mathcal{K}_{safe}$ represents the subset of stable aerodynamic flight dynamics bounded by internal collision-avoidance parameters. Set theory allows for the rigorous definition and handling of boundary conditions and edge cases. If an input $g \notin \mathcal{G}_{valid}$ is detected—for instance, if the performer drops their arm entirely or moves erratically beyond the sampling threshold—the mapping function strictly defaults to a null set operation, immediately triggering an autonomous, stable hover state in the drone.

Cybernetics: First-Order Control and Second-Order Coupling

The successful deployment of this architecture relies on nested tiers of cybernetic control loops. First-order cybernetics deals exclusively with observed systems, specifically the internal Proportional-Integral-Derivative (PID) control algorithms governing the drone's localized stabilization. The drone utilizes a Time-of-Flight (ToF) laser-ranging sensor to maintain a strict altitude setpoint along the Z-axis, completely independent of human capacitive input[15]. It closes its own internal feedback loop by measuring the error between the observed distance to the floor and the programmed setpoint, adjusting motor thrust autonomously to negate drift[15].
However, the complete performance ecosystem relies heavily on second-order cybernetics, a paradigm where the observer is structurally coupled to, and becomes an inseparable part of, the observed system[17]. In this architecture, the performer wearing the AR glasses is not an

external, detached controller operating a joystick, but a biological node deeply embedded within a continuous macro-feedback loop.

1. **Forward Path Execution:** The performer deliberately alters the electromagnetic field of the glove through physical motion. The capacitance-to-digital converter processes this perturbation, the localized microcontroller transmits the new kinematic setpoints via radio frequency, and the drone alters its physical position in space.
2. **Feedback Path Transmission:** The drone actively projects a localized hologram, while the AR glasses simultaneously overlay real-time DAW performance metrics (such as tempo, track EQ, and spatial grids) directly onto the performer's visual field.
3. **Biological Correction:** The performer visually observes the drone's actual position relative to the augmented holographic grid, processes this discrepancy cognitively, and dynamically adjusts their hand posture to correct the drone's placement.

This establishes a continuous, dynamic cybernetic equilibrium. The human brain acts as the macro-level predictive controller, organically compensating for any inherent network latency, sensor noise, or aerodynamic drift present in the drone's micro-level PID loops.

Hardware Architecture: The Glove and Drone Prototype

The immediate engineering objective is the fabrication of a singular control glove paired with a micro-drone. The hardware architecture prioritizes ultra-low latency, robust immunity to environmental noise, and high ergonomic flexibility for the performer.

The Capacitive E-Textile Glove

The glove serves as the primary sensor array, featuring seven distinct, localized control nodes: four embedded within the structure of the hand (positioned at the palm, thumb, index, and pinky regions) and three distributed sequentially up the arm (wrist, elbow, and shoulder).

Materials, Fabrication, and Shielding

The sensor nodes are fabricated using a specialized surface crochet technique, integrating highly conductive thread directly into a semi-open-mesh acrylic, bamboo, or faux fur yarn substrate[5]. This fabrication method yields sensors that are mechanically compliant, highly conformable to complex anatomical geometry, and easily modifiable for rapid prototyping[5].
For the primary conductive element, 316L grade stainless steel yarn is specified. While silver-plated threads are common in e-textiles, they are prone to oxidization which rapidly degrades long-term capacitive sensitivity[18]. The 316L stainless steel provides immense tensile strength and stable electrical resistivity, which is absolutely critical for maintaining a consistent baseline LC tank resonance[18]. Furthermore, to mitigate the capacitive and inductive crosstalk along the physical routing lines running up the arm, coaxial braided composite yarns are deployed. In these advanced structures, the active stainless steel core is surrounded by an insulating dielectric layer (such as polyamide) and a secondary grounded shield braid, preventing the transmission lines from acting as unintended antennas[19].

Sensor Interfacing: Texas Instruments FDC2214

The central data acquisition hardware is the Texas Instruments FDC2214, an industry-leading, high-resolution, 4-channel capacitance-to-digital converter[12]. Because a single FDC2214 supports a maximum of four channels, the 7-node glove architecture requires the deployment of two FDC2214 integrated circuits, communicating over a shared I2C bus to a localized microcontroller.

The FDC2214's innovative narrow-band LC resonator architecture provides unprecedented immunity to the severe electromagnetic interference (EMI) that plagues live electronic music environments, which are heavily saturated with radio frequencies, power supplies, and stage motors[11]. Each of the seven crocheted nodes is wired in parallel with a physical surface-mount inductor ($18\ \mu\mathrm{H}$) and a base capacitor ($33\ \mathrm{pF}$) to form the resonant LC tank, setting the baseline excitation frequency to approximately $13.7\ \mathrm{MHz}$[23].

The two FDC2214 chips are interfaced via I2C to an ESP32-S3 microcontroller mounted ergonomically on the wrist or forearm. The ESP32-S3 utilizes GPIO 1 (SCL) and GPIO 2 (SDA) for its default native I2C communication routing[25]. The ADDR pin on the second FDC2214 is physically pulled high to assign it a distinct, non-overlapping I2C address, preventing bus collisions during high-speed data polling[20].

Microcontroller and Network Operations

The ESP32-S3 serves as the master processing node for the entire arm, powered by a lightweight 3.7V LiPo battery. The ESP32-S3 was selected for its robust dual-core Xtensa LX7 architecture[28]. Core 0 is dedicated entirely to handling the high-speed I2C polling of the FDC2214 chips and executing the information-theoretic noise filtering algorithms, while Core 1 is isolated to handle the high-priority wireless transmission protocol, ensuring no blocking delays occur between sensor reading and packet transmission[27].

The Micro-Drone Platform

To safely hover in close proximity to a human performer and support eventual massive swarm deployment, the drone must be inherently lightweight, mechanically safe, and feature a highly open, programmable firmware architecture. The system utilizes a platform akin to the Crazyflie 2.1 ecosystem or a custom ESP32-S3-based micro-drone, such as the ESP-FLY[28].

Flight Controller and Spatial Sensors

The Crazyflie 2.1 relies on an STM32F405 processor and an onboard BMI088 6-axis inertial measurement unit (IMU) for localized self-leveling and stabilization[28]. The most critical hardware addition for this project is the Z-ranger deck, an expansion board featuring a downward-facing VL53L0x Time-of-Flight (ToF) laser-ranging sensor[15].

The purpose of the drone during this developmental stage is not fully unconstrained 3D manual flight, which is notoriously difficult to pilot via abstract gestures. Rather, the goal is localized hovering relative to the hand. The Z-ranger deck continuously measures the precise distance to the stage floor using a detection cone, allowing the drone's internal PID loops to maintain a strict altitude setpoint (e.g., $40\ \mathrm{cm}$)[15]. Consequently, the complex inputs from the glove are mapped strictly to planar velocities (X and Y axes) and yaw, drastically reducing the cognitive load on the performer and preventing catastrophic altitude drops during rapid musical movements.

Communication Layer: The ESP-NOW Protocol

In live performance environments, standard 802.11 Wi-Fi networks are prone to unacceptable latency spikes, packet loss, and router dropouts due to audience device congestion. To achieve the rigorous timing required for fluid cybernetic feedback, the system bypasses the traditional Wi-Fi networking stack entirely by utilizing ESP-NOW[31].

ESP-NOW is a proprietary, connectionless, peer-to-peer communication protocol developed by Espressif that operates directly on the data-link layer. By stripping the OSI model down to a single layer, it achieves extraordinary millisecond-level response times, typically reporting latency in the range of 7 to $25\ \mathrm{ms}$ [28]. The ESP32-S3 on the glove acts as the ESP-NOW broadcaster, transmitting 256-bit AES-GCM encrypted payload packets directly to the MAC address of the drone's receiving module. This receiving module can either be a secondary ESP32 bridged to the Crazyflie flight controller via UART[33], or directly to the primary flight controller if utilizing the fully integrated ESP-FLY architecture[28].

Algorithmic Control and Data Processing Pipeline

The delicate transition from raw, fluctuating analog capacitance into stable, aerodynamic drone flight requires a highly deterministic, multi-stage software pipeline executed on the ESP32-S3.

FDC2214 Configuration and Calibration Routines

To achieve the theoretical stable noise floor of $\sim 0.3\ \mathrm{fF}$ [12], the internal registers of the FDC2214 must be meticulously configured via I2C upon boot. The sampling rates and deglitch filters must be optimized for the relatively slow speeds of human gesture articulation (typically sub- $50\ \mathrm{Hz}$) rather than high-frequency mechanical vibration[24].

Register Name	Hex Address	Configuration Rationale & Technical Settings
RCOUNT_CHx	0x08 - 0x0B	Defines the reference count for the conversion time. Configured to budget a conversion window of

		approximately $3.33\ \mathrm{ms}$ per channel. This ensures high-resolution 28-bit reads while allowing rapid multiplexed switching across all four channels on the IC[21].
SETTLECOUNT_CHx	0x10 - 0x13	Determines the precise LC tank settling time before the measurement cycle begins. Set to a minimum hex value of 0x000A (10) to allow the $13.7\ \mathrm{MHz}$ oscillation to stabilize, yielding a brief $4\ \mu s$ delay[21].
CLOCK_DIVIDER_CHx	0x14 - 0x17	Sets the reference oscillator frequency. Configured specifically to utilize the external $40\ \mathrm{MHz}$ crystal oscillator (Y2) for maximum thermal stability, bypassing the less accurate internal oscillator[20].
CONFIG	0x1A	The primary operational configuration register. Activates continuous multi-channel scanning mode and enables the INTB pin to trigger hardware interrupts on the ESP32-S3 precisely when new data is ready for extraction[21].
MUX_CONFIG	0x1B	Configures the input deglitch filter to suppress high-frequency EMI and

		ringing. Set to the lowest possible bandwidth that safely exceeds the maximum baseline LC frequency of the sensors[36].

Upon initialization of the system, the ESP32-S3 executes a mandatory calibration routine. The performer must hold their arm in a neutral, relaxed state, distanced from the drone and other conductive bodies. The system rapidly polls and averages 1,000 samples per node to establish a highly accurate baseline capacitance C_{base_i} for each node $i \in \{1, 2, \dots, 7\}$.

Filtering and Mapping to Flight Dynamics

During live musical operation, the raw capacitance delta $\Delta C_i = C_{current_i} - C_{base_i}$ is extracted continuously. The data processing pipeline proceeds through the following sequence:

1. **Signal Smoothing:** A one-dimensional Kalman filter or a low-pass Exponential Moving Average (EMA) is applied to the raw ΔC_i stream to aggressively eliminate high-frequency jitter and sensor noise.
2. **Crosstalk Matrixing:** Because extreme anatomical movements (e.g., deeply bending the elbow) may inadvertently alter the proximal shoulder node's capacitance, the predefined information theory framework is applied. A calibrated transfer matrix subtracts the mapped mutual capacitance overlap, mathematically isolating the distinct node activations.
3. **Thresholding and State Machine Logic:** A variable hysteresis threshold algorithm determines if a specific node is actively engaged, preventing minor twitches from triggering drone movement.
4. **Kinematic Translation:** The isolated, active node states are mapped directly to flight vectors using the drone's Python API equivalent command structures (e.g., send_hover_setpoint(vx, vy, yawrate, zdistance))[40].

For the prototype stage, the mapping topology is configured as follows:

- **Hand Nodes (4):** Modulate the horizontal planar velocity (V_x, V_y). Tilting the palm forward or backward increases V_x, allowing the drone to be visually "pushed" or "pulled" by the hand's electric field[41].
- **Wrist Node:** Modulates the yaw rate. Rotating the wrist along its axis spins the drone in place, a crucial movement for aiming the onboard holographic projector.
- **Elbow/Shoulder Nodes:** Modulate the Z-distance setpoint. While the drone autonomously holds its height, raising the elbow adjusts the Z-ranger's target variable from the baseline 40 cm up to 80 cm or higher, elevating the drone in seamless

synchronization with the performer's body[15].

The resulting compiled setpoint packet is transmitted over the ESP-NOW network at regular intervals of 20 ms. This rigorous transmission rate is required to satisfy the drone's internal safety protocols; if the flight controller does not receive an updated setpoint within a brief timeout window, an automatic shutdown is triggered, and the drone will drop from the sky to prevent uncontrolled flyaways[40].

Future Scalability Roadmap: The Full Cyber-Physical Performance Suit

While the immediate prototype establishes the aerodynamic and computational viability of a 7-node arm controlling a single hovering drone, the ultimate end-state of this project is a sweeping cyber-physical performance interface. The roadmap culminates in a full-body fabric suit controlling a decentralized swarm of thirteen micro-drones, heavily integrated with a custom Mixxx DAW and AR glasses for live electronic music manipulation.

The 13-Drone Holographic Swarm Architecture

Scaling the drone count from one to thirteen requires transitioning the network topology from a single peer-to-peer link to a broadcast or sophisticated multi-peer star topology[29]. Fortunately, the ESP-NOW protocol is highly scalable, supporting multiple peers within a single network with virtually no penalty to latency, bounded primarily by channel capacity and processing power[29].

Crucially, the thirteen drones will not be individually micromanaged by the performer; attempting to do so would instantly exceed human cognitive capacity and result in swarm collisions. Instead, the drones will utilize advanced distributed swarm logic[33]. The full-body suit will broadcast macroscopic, high-level setpoints (e.g., "Orbit center," "Expand radius," "Cascade elevation"). The master flight controller will translate the human capacitive gestures into these macro-commands, while the individual drones rely on their onboard multi-directional ToF sensors (such as the Multi-ranger deck) to maintain relative spatial formations and prevent mid-air collisions[30].

Furthermore, the integration of lightweight holographic projectors fundamentally alters the visual nature of the performance. The suit's gestures will simultaneously control the audio output of the DAW (via MIDI over Wi-Fi/ESP-NOW) and modulate the holographic visual parameters. For example, a sharp, sudden wrist flick could trigger a heavy bass drop within the Mixxx software while concurrently commanding the drone swarm to pulse their holographic projections in exact synchronization with the beat.

Full-Body Sensor Topology and Tri-Functional Nodes

The scalability of the fabric suit requires replicating the proven 7-node modular architecture across the entire human body. The proposed final topology yields 42 total nodes, segmented logically into distinct anatomical sub-systems.

Anatomical Segment	Node Count	Specific Node Placement	Primary Cybernetic Function
Right Arm	7	4x Hand, Wrist, Elbow, Shoulder	Swarm spatial manipulation (X/Y plane) and Mixxx track 1 EQ sweeping.
Left Arm	7	4x Hand, Wrist, Elbow, Shoulder	Hologram parameter control, Swarm Yaw orientation, and Mixxx track 2 EQ.
Right Leg	7	4x Footwear, Ankle, Knee, Hip	Global tempo control, kick drum MIDI triggers, swarm altitude baseline.
Left Leg	7	4x Footwear, Ankle, Knee, Hip	DAW effect racks (reverb/delay wetness), localized floor-level FX.
Body/Torso	7	Chest, Back, Obliques, Spine	Core suit network connection, master audio volume, macro-swarm formations.
Head/AR Interface	7	Horn/Ear side nodes, AR Glasses	AR HUD navigation, visual feedback looping, drone targeting validation.

Note: Each of these 7-node clusters operates as a mathematically independent sub-system,

utilizing its own dedicated ESP32-S3 and a paired set of FDC2214 CDCs to ensure no single point of processing failure brings down the entire suit.
To manage this massive influx of 42 distinct 28-bit data streams, the suit will employ a robust I2C multi-master architecture, or more likely, a CAN bus network routing all peripheral ESP32-S3 sub-controllers to a high-powered central processing unit located on the torso. This central hub will execute complex sensor fusion algorithms, translating the massive 42-dimensional capacitance vector into a unified, lightweight JSON or OSC (Open Sound Control) packet for the DAW and the drone swarm.
The crocheted nodes throughout the suit will be explicitly engineered to be "tri-functional," fulfilling three simultaneous system requirements:

1. **Suit Connection:** Establishing the internal data and power routing backbone of the garment, utilizing the braided shielding of the yarn to act as a physical and electrical tether[19].
2. **Drone Connection:** Modulating the external RF fields. The nodes act as the physical trigger mechanisms that dictate the generation of ESP-NOW packets, directly commanding the external swarm.
3. **Localized Limb Control:** Managing specific parameter constraints relative only to that anatomical segment's physical limits, ensuring that an errant leg movement does not accidentally trigger an arm-based spatial command.

By fully realizing this tri-functional, 42-node architecture, the system transcends a simple wearable controller, becoming a fully immersive cyber-physical instrument that blurs the boundary between the performer, the music, the visual field, and the autonomous drone swarm.

Long-Term Goal: Digital, AR, and Holographic Visual Interface Progression

- Phased visual interface deployment prioritizing Augmented Reality (AR) glasses and digital camera overlays as the foundational first step before physical volumetric/holographic projection.
- Digital presence rendering: Rendering agent and visual augmentations digitally across AR headsets, camera feeds, and connected digital/cybernetic displays.
- Expressive and perceptual interface mechanics: Integrating visual representations and gestures with the agent architecture and performance system.

Works cited

1. (2001..2003) History of Pen, Touch and Gesture Computing, http://users.erols.com/rwservices/pens/biblio03.html
2. Theremin - Physics Lecture Demos, https://lecturedemos.ph.utexas.edu/demo/313
3. C64 Theremin, https://www.linusakesson.net/hardware/theremin/
4. How a Theremin Works - MicroKits, https://microkits.net/pages/how-a-theremin-works

5. Crocheted Capacitive Touch Sensors for Rapid Prototyping of Soft Interfaces - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC13241003/
6. Crocheted Capacitive Touch Sensors for Rapid Prototyping of Soft Interfaces - PubMed, https://pubmed.ncbi.nlm.nih.gov/42253449/
7. Casimir effect - Wikipedia, https://en.wikipedia.org/wiki/Casimir_effect
8. NEW DEVELOPMENTS IN THE CASIMIR EFFECT, https://web.pa.msu.edu/courses/2006spring/AST860/papers/BordagMohideenMostepanenenko2001.pdf
9. (PDF) The Casimir Effect in the QRG Framework - ResearchGate, https://www.researchgate.net/publication/408523063_The_Casimir_Effect_in_the_QRG_Framework
10. Exact Multimode Quantization of Superconducting Circuits via Boundary Admittance - arXiv, https://arxiv.org/html/2601.04407v1
11. Capacitive Frost or Ice Detection Reference Design—Resolution of < 1 mm - Texas Instruments, https://www.ti.com/lit/ug/tidud79/tidud79.pdf
12. FDC2214 data sheet, product information and support | TI.com, https://www.ti.com/product/FDC2214
13. (PDF) Woven Fabric-Based Electrical Circuits: Part II: Yarn and Fabric Structures to Reduce Crosstalk Noise in Woven Fabric-Based Circuits - ResearchGate, https://www.researchgate.net/publication/233854923_Woven_Fabric-Based_Electrical_Circuits_Part_II_Yarn_and_Fabric_Structures_to_Reduce_Crosstalk_Noise_in_Woven_Fabric-Based_Circuits
14. University of Nevada, Reno Multimodal and Interactive ... - ScholarWolf, https://scholarwolf.unr.edu/bitstreams/dd95276b-b009-411c-865b-974c6f1d00d3/download
15. Getting started with the Z-ranger deck - Bitcraze, https://www.bitcraze.io/getting-started-with-z-ranger-deck/
16. Autonomous flight of the micro drone Crazyflie 2.1 through an obstacle course - DiVA Portal, https://www.diva-portal.org/smash/get/diva2:1440184/FULLTEXT01.pdf
17. Designing Academic Conferences in the Light of Second-Order, https://constructivist.info/articles/11/1/065.richards.pdf
18. Conductive Thread Mastery: Sewing Techniques, Applications, and Expert - MaggieFrames, https://www.maggieframes.com/blogs/embroidery-blogs/conductive-thread-mastery-sewing-techniques-applications-and-expert-tips
19. US20220056619A1 - Functional Braided Composite Yarn - Google Patents, https://patents.google.com/patent/US20220056619A1/en
20. Texas Instruments FDC2214 Capacitance to Digital - Review - element14 Community, https://community.element14.com/products/roadtest/rv/roadtest_reviews/557/texas-instruments-fdc2214-capacitance-to-digital-review
21. FDC2x1x Multi-Channel, High Resolution Capacitance-to-Digital Converter for Capacitive Sensing Applications datasheet (Rev. B) - Texas Instruments, https://www.ti.com/lit/gpn/FDC2114

22. TI revolutionizes capacitive sensing with world's only noise-immune solution - PR Newswire, https://www.prnewswire.com/news-releases/ti-revolutionizes-capacitive-sensing-with-worlds-only-noise-immune-solution-300161698.html
23. A Novel Adaptive Flexible Capacitive Sensor for Accurate Intravenous Fluid Monitoring in Clinical Settings - PMC, https://pmc.ncbi.nlm.nih.gov/articles/PMC12299985/
24. Unimodal and Multimodal Sensor Fusion for Wearable Activity Recognition - kluedo, https://kluedo.ub.rptu.de/files/8585/_PhDThesis-Bello.pdf
25. Example Code for Arduino - ESP32-S3 native I2C interface - DFRobot Wiki, https://wiki.dfrobot.com/dfr0944/docs/23781
26. ESP32-S3 DevKitC Pinout Reference Guide: GPIOs Explained - Random Nerd Tutorials, https://randomnerdtutorials.com/esp32-s3-devkitc-pinout-guide/
27. Inter-Integrated Circuit (I2C) - ESP32-S3 - — ESP-IDF Programming Guide v6.0.2 documentation - Espressif Systems, https://docs.espressif.com/projects/esp-idf/en/stable/esp32s3/api-reference/peripherals/i2c.html
28. ESP-FLY micro drone kit offers ESP32-S3-based flight control and ESP-NOW support, https://linuxgizmos.com/esp-fly-micro-drone-kit-offers-esp32-s3-based-flight-control-and-esp-now-support/
29. Using ESP-NOW in Arduino - Espressif Developer Portal, https://developer.espressif.com/blog/2024/08/arduino-esp-now-lib/
30. Crazyflie 2.1 | PX4 Guide (main), https://docs.px4.io/main/en/complete_vehicles_mc/crazyflie21
31. ESP-NOW Tutorial for Long-Range Communication: Building Robust Wireless Networks in 2025 - Think Robotics, https://thinkrobotics.com/blogs/learn/esp-now-tutorial-for-long-range-communication-building-robust-wireless-networks-in-2025
32. I built a motion capture system which sends data to drones using ESP32s & ESP NOW so they can fly autonomously - details in comments! - Reddit, https://www.reddit.com/r/esp32/comments/17xj326/i_built_a_motion_capture_system_which_sends_data/
33. DroneBridge for ESP32 — Plane documentation - ArduPilot, https://ardupilot.org/plane/docs/common-esp32-telemetry.html
34. DroneBridge for ESP32. A secure & transparent telemetry link with support for WiFi and ESP-NOW. Supporting MAVLink, MSP, LTM or any other protocol - GitHub, https://github.com/DroneBridge/ESP32
35. Overview - DroneBridge Docs, https://dronebridge.gitbook.io/docs/dronebridge-for-esp32/untitled
36. LDC1312-Q1 Datasheet | PDF | Electrostatic Discharge - Scribd, https://www.scribd.com/document/703972606/LDC1312-Q1-Datasheet
37. Mk Fdc2214 技术手册 - Scribd, https://www.scribd.com/document/582605529/Mk-Fdc2214-%E6%8A%80%E6%9C%AF%E6%89%8B%E5%86%8C

38. Exploration of planar capaciflector for wearable respiration monitoring - ePrints Soton - University of Southampton, https://eprints.soton.ac.uk/496508/1/Boyuan_Zhao_PhD_Thesis_final_version_2024.pdf
39. LDC1612/LDC1614 Inductive Sensors Datasheet | PDF | Manufactured Goods - Scribd, https://www.scribd.com/document/662540840/ldc1612
40. The Commander Framework part 2: offboard or onboard? - Bitcraze, https://www.bitcraze.io/2024/01/the-commander-framework-part-2-offboard-or-onboard/
41. Flightcontrol Tab - Bitcraze, https://www.bitcraze.io/documentation/repository/crazyflie-clients-python/master/userguides/userguide_client/flightcontrol_tab/
42. autonomy - Bitcraze, https://www.bitcraze.io/tag/autonomy/